tb_cocotb.v

AUTHORS

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DATES

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INFORMATION

Brief

Test bench wrapper for cocotb

License MIT

tb_cocotb

```
module tb_cocotb #(
parameter
integer
ADDRESS_WIDTH
 =
32,
parameter
integer
BUS_WIDTH
 =
4,
parameter
 [ADDRESS_WIDTH-1:0]
SLAVE_ADDRESS
```

```
 =
32'h44A20000,
parameter
 [ADDRESS_WIDTH-1:0]
SLAVE_REGION
 =
32'h0000FFFF
)
                                                                         (
output
wire
connected,
input
wire
aclk,
input
wire
arstn,
input
wire
 [ADDRESS_WIDTH-1:0]
s_axi_araddr,
input
wire
 [2:0]
s_axi_arprot,
input
wire
s_axi_arvalid,
output
wire
s_axi_arready,
output
wire
 [BUS_WIDTH*8-1:0]
s_axi_rdata,
output
wire
 [1:0]
s_axi_rresp,
output
wire
s_axi_rvalid,
input
wire
s_axi_rready,
output
wire
 [ADDRESS_WIDTH-1:0]
m_axi_araddr,
output
wire
 [2:0]
m_axi_arprot,
output
wire
m_axi_arvalid,
input
wire
m_axi_arready,
input
wire
 [BUS_WIDTH*8-1:0]
m_axi_rdata,
input
wire
```

```
 [1:0]
m_axi_rresp,
input
wire
m_axi_rvalid,
output
wire
m_axi_rready,
input
wire
 [ADDRESS_WIDTH-1:0]
s_axi_awaddr,
input
wire
 [2:0]
s_axi_awprot,
input
wire
s_axi_awvalid,
output
wire
s_axi_awready,
input
wire
 [BUS_WIDTH*8-1:0]
s_axi_wdata,
input
wire
 [BUS_WIDTH-1:0]
s_axi_wstrb,
input
wire
s_axi_wvalid,
output
wire
s_axi_wready,
output
wire
 [1:0]
s_axi_bresp,
output
wire
s_axi_bvalid,
input
wire
s_axi_bready,
output
wire
 [ADDRESS_WIDTH-1:0]
m_axi_awaddr,
output
wire
 [2:0]
m_axi_awprot,
output
wire
m_axi_awvalid,
input
wire
m_axi_awready,
output
wire
 [BUS_WIDTH*8-1:0]
m_axi_wdata,
output
wire
```

```
  [BUS_WIDTH-1:0]
 m_axi_wstrb,
 output
 wire
 m_axi_wvalid,
 input
 wire
 m_axi_wready,
 input
 wire
  [1:0]
 m_axi_bresp,
 input
 wire
 m_axi_bvalid,
 output
 wire
 m_axi_bready
)
```

AXI lite address recoder for read channel. Write channel lines included for cocotb cores to connect to.

Parameters

ADDRESS_WIDTH `parameter integer`	Width of the AXI LITE address port in bits.
BUS_WIDTH `parameter integer`	Width of the AXI LITE bus data port in bytes.
SLAVE_ADDRESS `parameter [ADDRESS_WIDTH- 1:0]`	Array of Addresses for each slave (0 = slave 0 and so on).
SLAVE_REGION `parameter [ADDRESS_WIDTH- 1:0]`	Region for the address that is valid for the SLAVE ADDRESS.

Ports

connected `output wire`	Core has established channel connection
aclk `input wire`	Input clock
arstn `input wire`	Input negative reset
s_axi_araddr `input wire [ADDRESS_WIDTH- 1:0]`	Slave read chanel input address.
s_axi_arprot `input wire [2:0]`	Slave read chanel input address protection
s_axi_arvalid `input wire`	Slave read chanel input address is valid
s_axi_arready `output wire`	Slave read chanel input is ready.
s_axi_rdata `output wire [BUS_WIDTH* 8- 1:0]`	Slave read chanel input data.
s_axi_rresp `output wire [1:0]`	Slave read chanel input data response.
s_axi_rvalid `output wire`	Slave read chanel input data valid
s_axi_rready `input wire`	Slave read chanel input is ready.
m_axi_araddr `output wire [ADDRESS_WIDTH- 1:0]`	Master read chanel output address.

m_axi_arprot
`output wire [2:0]`

Master read chanel output address protection

m_axi_arvalid
`output wire`

Master read chanel output address is valid

m_axi_arready
`input wire`

Master read chanel output is ready.

m_axi_rdata
`input wire [BUS_WIDTH* 8- 1:0]`

Master read chanel output data.

m_axi_rresp
`input wire [1:0]`

Master read chanel output data response.

m_axi_rvalid
`input wire`

Master read chanel output data valid

m_axi_rready
`output wire`

Master read chanel output is ready.

INSTANTIATED MODULES

dut

Device under test, axi_lite_read_channel_decoder